

Activity 3: LLM Limitations

TASK 1: Solving a complex mathematical question.

Prompt:

Solve this mathematical question:

a, b, and c are consecutive terms in an arithmetic progression.

a = 4 and c = 20.

a, c, and d are consecutive terms in a geometric progression

Find b + d

LLM Response:

To solve this problem, we need to first understand the properties of arithmetic and geometric progressions.

In an arithmetic progression, each term is obtained by adding a constant difference to the previous term. Since a, b, and c are consecutive terms in an arithmetic progression, we can write:

$$b = a + d$$

$$c = a + 2d$$

Given that a = 4, we can substitute this value into the equations:

$$b = 4 + d$$

$$c = 4 + 2d$$

We are also given that c = 20. We can set up an equation using this information:

$$4 + 2d = 20$$

Subtracting 4 from both sides gives:

$$2d = 16$$

Dividing both sides by 2 gives:

$$d = 8$$

Now that we have found d, we can find b and d:

$$b = 4 + d$$

$$b = 4 + 8$$

$$b = 12$$

$$d = 8$$

So, $b = 12$ and $d = 8$.

To find the sum of b and d , we can simply add these two values together:

$$b + d = 12 + 8$$

$$b + d = 20$$

Therefore, the sum of b and d is 20.

TASK 2: Probing for factual knowledge in a niche area

Prompt:

In UK corporation tax reducing balance capital allowances, what is the current main pool rate?

LLM Response:

In the UK, the main pool rate for tax reducing balance capital allowances is 90% of the gain. This rate is typically used for tax reducing balance (TRB) relief. The current main pool rate for TRB relief is 90% of the gain, which is £1,884.40 in 2022-23.

Prompt:

How many restaurants can be found on Great Horton Road in Bradford, England?

LLM Response:

I couldn't find any specific information on the number of restaurants on Great Horton Road in Bradford, England. However, I can suggest some possible sources where you might be able to find the information you're looking for:

* Google Maps: You can use Google Maps to view the location and number of restaurants along Great Horton Road. Simply enter the address and zoom in on the area to see the list of restaurants.

* Online Directories: Websites like TripAdvisor, Yelp, or Zomato may have a list of restaurants on Great Horton Road. These websites often provide information on the number of restaurants in an area.

* Local Government Website: The City of Bradford Council's website may have information on the number of restaurants on Great Horton Road, as well as other information about the area.

If you have any more information or context, I may be able to help you find the specific information you're looking for.